

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

Centre of Excellence in Education

BIOLOGY

Life Science

6010/3

NOVEMBER 2023

1 hour 30 minutes

Write your answers in the spaces provided on the question paper.

TIME: 1 hour 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the question paper.

Answer any Five questions.

Write your answers in the spaces provided on the question paper.

The first three pages of the question paper are for the questions you have attempted.

If you have more than one sheet of paper, insert the sheets together in the rest of the examination.

INFORMATION FOR CANDIDATES

The number of marks for each question is given in brackets () at the end of each question on the question paper.

Use a pencil to write your answers. Write in good English and show your working in your answers.

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- a) Describe the molecular structure of DNA. [8]
- b) Suggest employment benefits which can be realised from conservation of biodiversity. [6]
- c) Explain, using a named example, the effects of gene mutation on an organism. [6]

- 1 (a) Describe the molecular structure of starch with reference to: [5]
 (i) amylose [3]
 (ii) amylopectin [6]
 (b) Explain how pH affects the rate of an enzyme-catalysed reaction. [6]
 (c) Explain the function of mRNA in protein synthesis. [6]
- 2 (a) Describe [2]
 (i) metaphase of mitosis. [6]
 (ii) telophase of mitosis. [6]
 (b) Explain, using **one** example, how the environment may affect the phenotype of an organism. [6]
 (c) Explain the use of artificial selection in agriculture. [6]
- 3 (a) Outline the functions of lipids in living organisms. [8]
 (b) Compare anaerobic respiration in mammalian cells to anaerobic respiration in yeast cells. [6]
 (c) Explain the importance of the link reaction in aerobic respiration. [6]
- 4 (a) Outline any **eight** uses of ATP in living organisms. [8]
 (b) Explain the production of ATP by non-cyclic photophosphorylation. [6]
 (c) Relate the structure of the chloroplast to its functions. [6]
- 5 (a) Describe the transcription of DNA to mRNA. [8]
 (b) Explain the global distribution of
 (i) tuberculosis (TB), [6]
 (ii) coronary heart disease (CHD). [6]
- 6 (a) Outline the economic importance of Kingdom Prokaryotae. [8]
 (b) Explain any **three** roles of nucleotides in living organisms. [6]
 (c) Relate the structure of the nucleus to its function. [6]